

Towards Geospatial Embeddings

Investigating Accurate and Accessible Deep Geospatial Feature Representations

Konstantin Klemmer · Anthony Fuller · Evan Shelhamer · Esther Rolf · Marc Rußwurm

$$E : (\text{lon}, \text{lat}, \text{time}) \rightarrow \text{emb} \in \mathbb{R}^d$$

Two teaching blocks, two hands-on demos

08:30

Teach 1

AI for EO 101 & Applications — what embeddings are and what they unlock

09:15

Demo 1

Using Earth embeddings: direct prediction

09:45

Break

10:10

Teach 2

Types of Earth embeddings: explicit vs implicit (INR)

10:45

Demo 2

Make your own embeddings: MOSAICS RCF & similarity maps

11:30

Close

Discussion, resources & Q&A

This deck covers the two teaching blocks (highlighted). Demos are run live.

Three learning outcomes

LO 1

Understand

What an *Earth embedding* is, and where it sits in the AI for Earth observation landscape.

LO 2

Apply

Use *Earth embeddings* in a real workflow: for mapping, as geographic priors, and for fast global search & retrieval.

LO 3

Distinguish

Tell explicit (pre-computed) from implicit (INR) *Earth embeddings* – and learn how to choose the right one for your use case.

PART 1

AI for Earth Observation 101

Foundation models, embeddings, and a new kind of remote sensing data

Presented by **Anthony Fuller**

$$E : (\text{lon}, \text{lat}, \text{time}) \rightarrow \text{emb} \in \mathbb{R}^d$$

We are drowning in Earth data

~800 PB

of Earth observation data exists
today

~4 TB

new Sentinel-2 imagery, every
single day

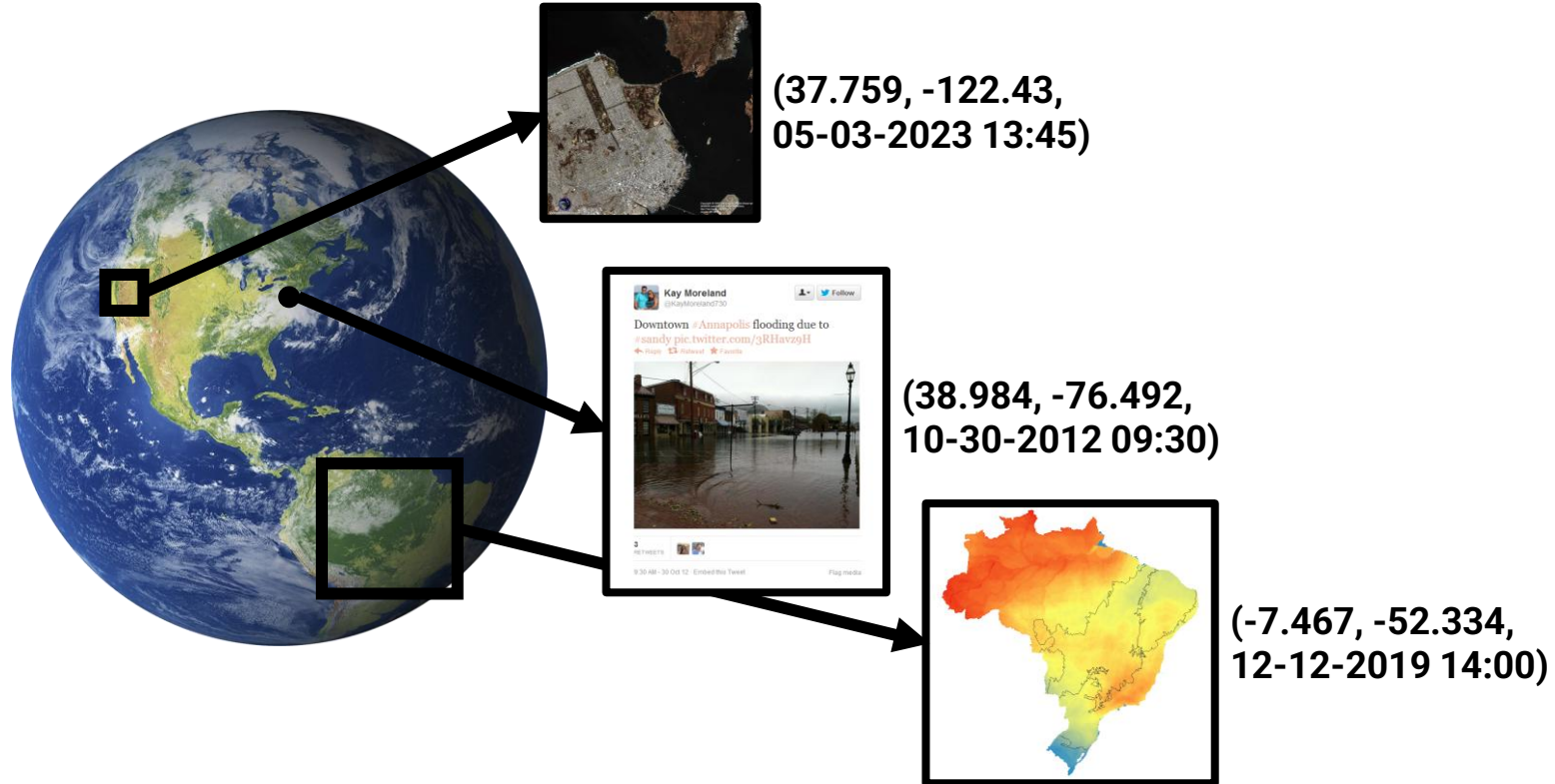
100s

of satellites continuously
imaging the planet

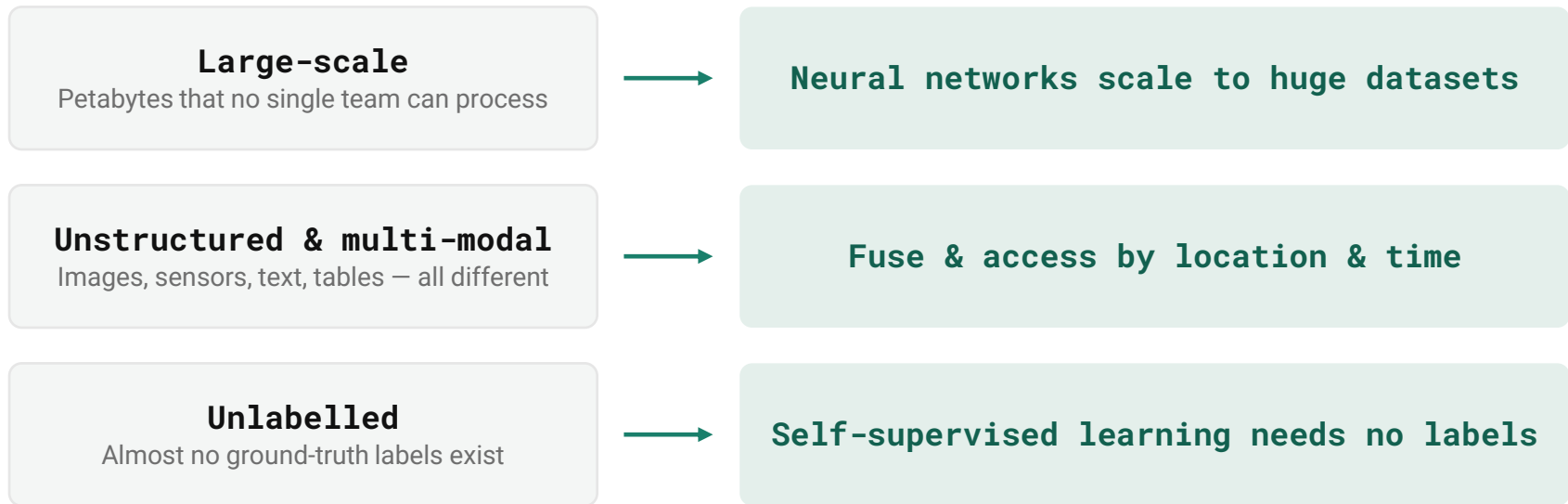
For scale: GPT-4 was trained on roughly 1 PB of text tokens. Earth observation data is ~800× larger — and almost none of it is labelled.



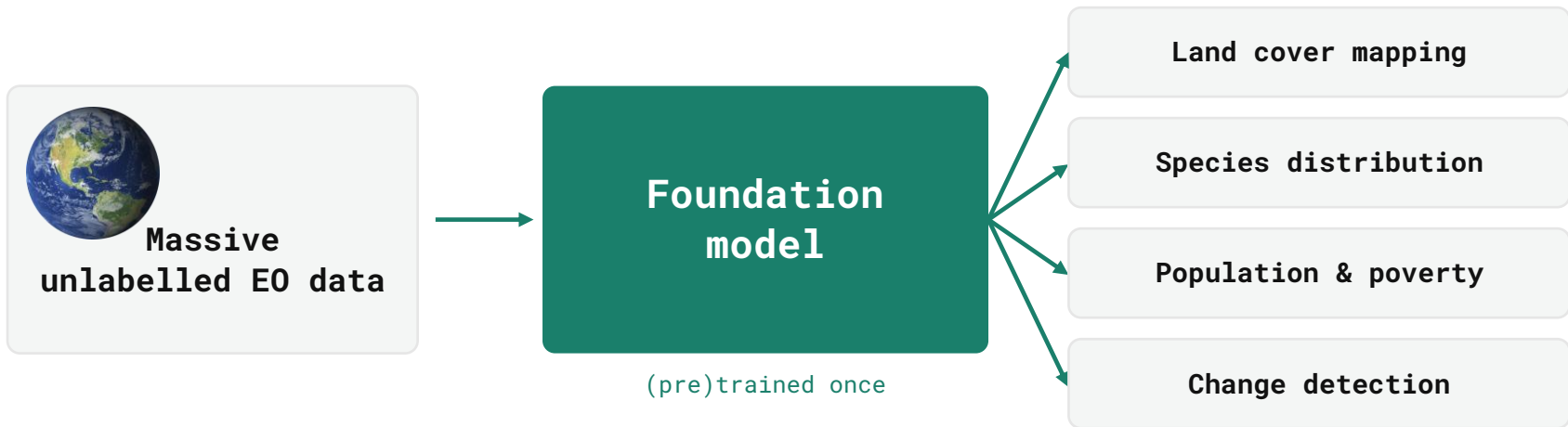
All geospatial data shares one index



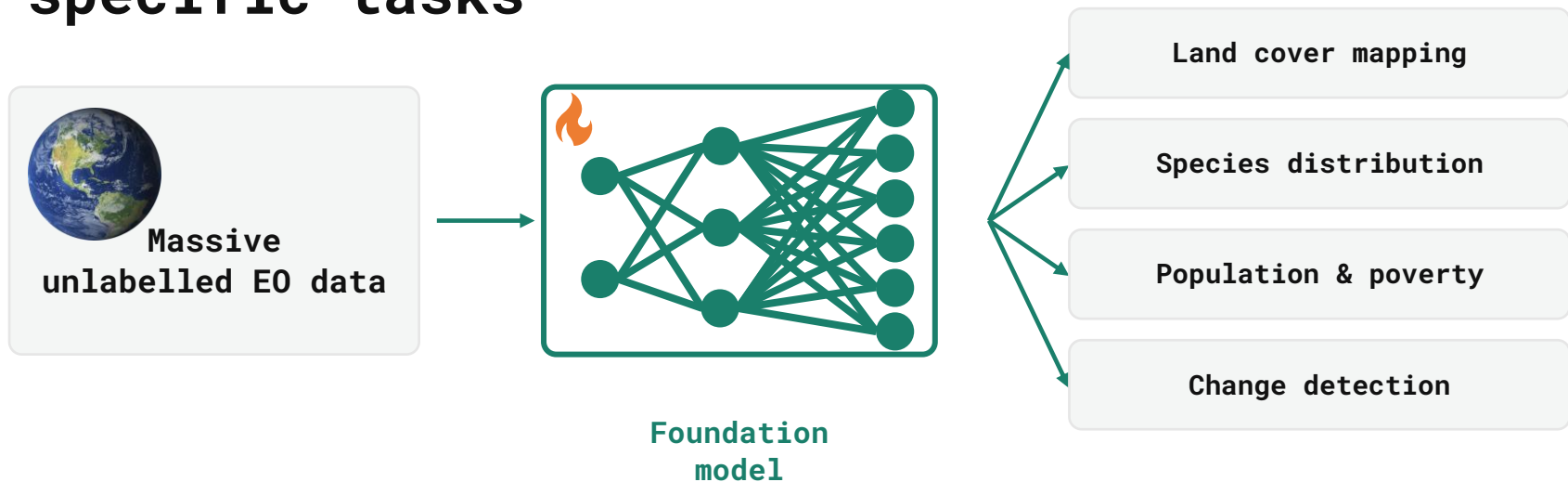
How AI unlocks this data



What is a foundation model?

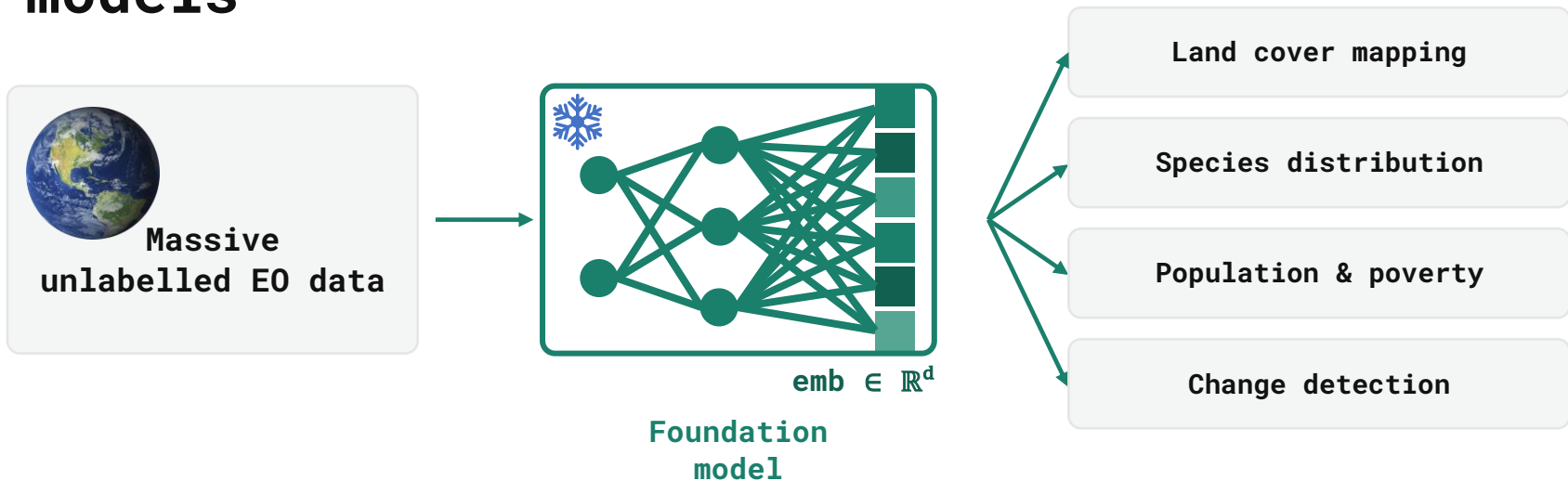


Foundation models can be fine-tuned for specific tasks



Continue training the foundation model – using data and labels from your specific task!

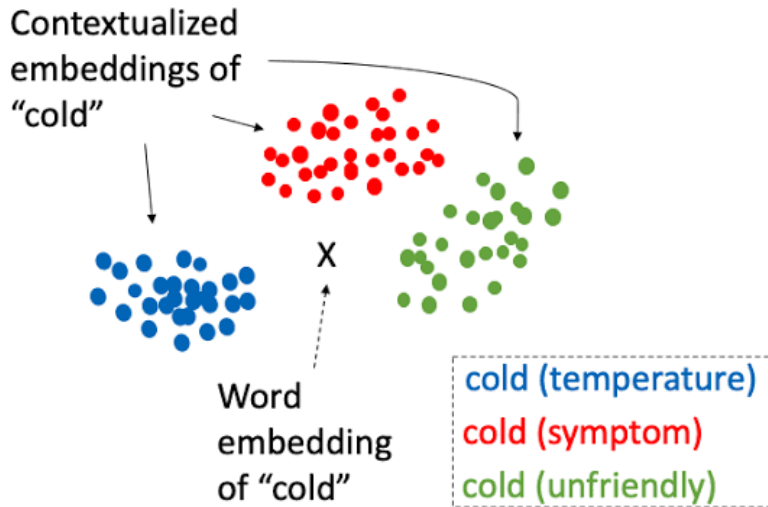
General purpose embeddings from foundation models



Embeddings are “general purpose” representations learned by foundation models. You can use them directly to predict your outcome – without the need for expensive re-training!

What is an embedding?

An embedding is a vector that places similar things close together and different things far apart.



- **Distance = similarity.** Nearby vectors mean similar places.
- **Comparable.** Cosine similarity gives a single number.
- **Reusable.** One vector feeds many downstream tasks.

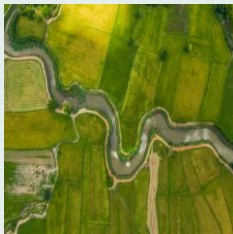
Embeddings revolutionised language

king - man + woman $\approx$ queen

- **Word2Vec & GloVe captured meaning, not spelling.** Words used in similar contexts get similar vectors.
- **The same trick powers today's large language models.** “Distance between words” is the foundation of modern NLP.
- **Earth embeddings bring this to geographic places and times.** Places that look and behave alike land close together.

Functional, not geographic, similarity

road + river $\approx$ bridge



Embeddings also vary over time:

- **Equatorial Costa Rica:** summer and winter embeddings look nearly identical.
- **Highly seasonal Alaska:** summer and winter differ strongly — a built-in signal of change.

Earth embeddings as learned “measurements”

SENSED measurement

What an instrument physically records — reflectance in each spectral band.

[0.07, 0.12, 0.31, ...] bands

Raw, large, sensor-specific — interpretable in the sense that each number corresponds to a band

LEARNED “measurement”

What a model learns to represent — a vector that captures the character of a place.

 $\text{emb} \in \mathbb{R}^d$

Compact, comparable, reusable across many tasks — not directly interpretable

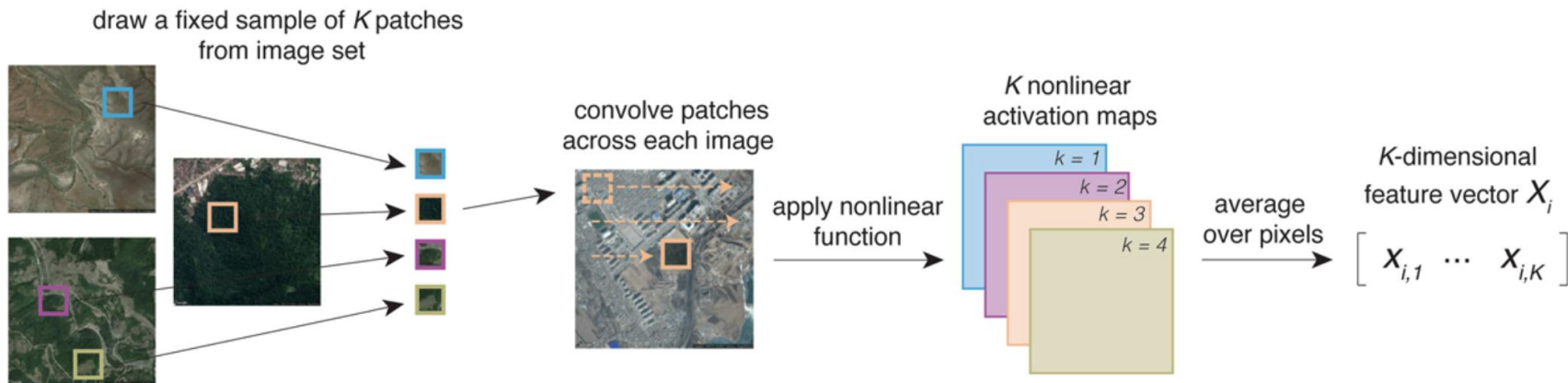
Same goal as remote sensing — describe a location — but the description is learned, not directly sensed.

From image to embedding: MOSAIKS



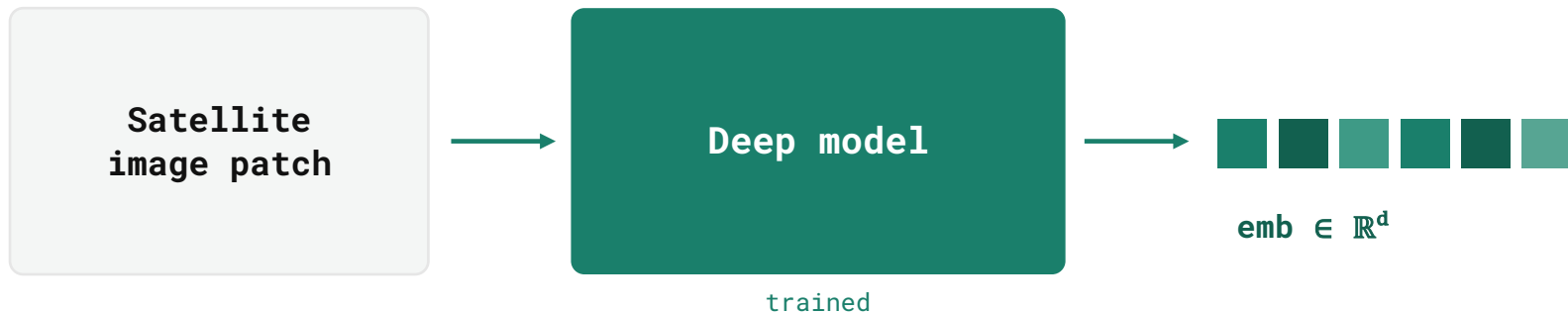
MOSAIKS shows even random features turn imagery into vectors that predict forest cover, income, population and more — with a simple linear model on top.

From image to embedding: MOSAIKS



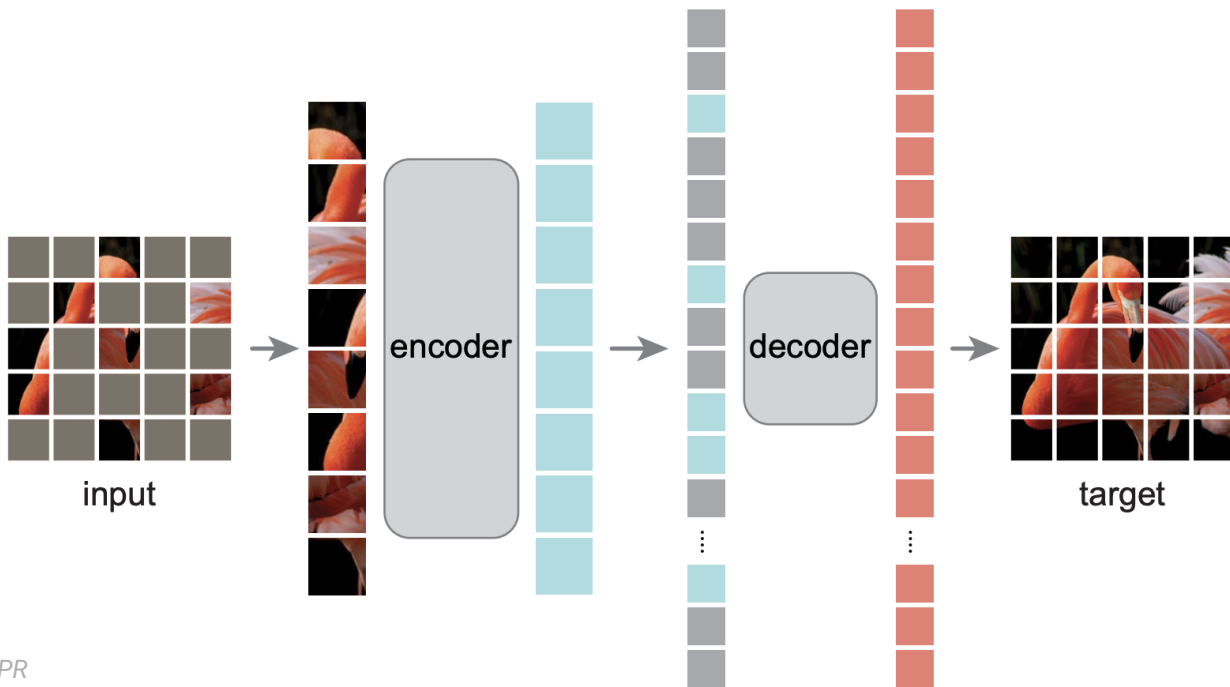
MOSAIKS shows even random features turn imagery into vectors that predict forest cover, income, population and more – with a simple linear model on top.

Learn features instead of designing them (or picking them at random)

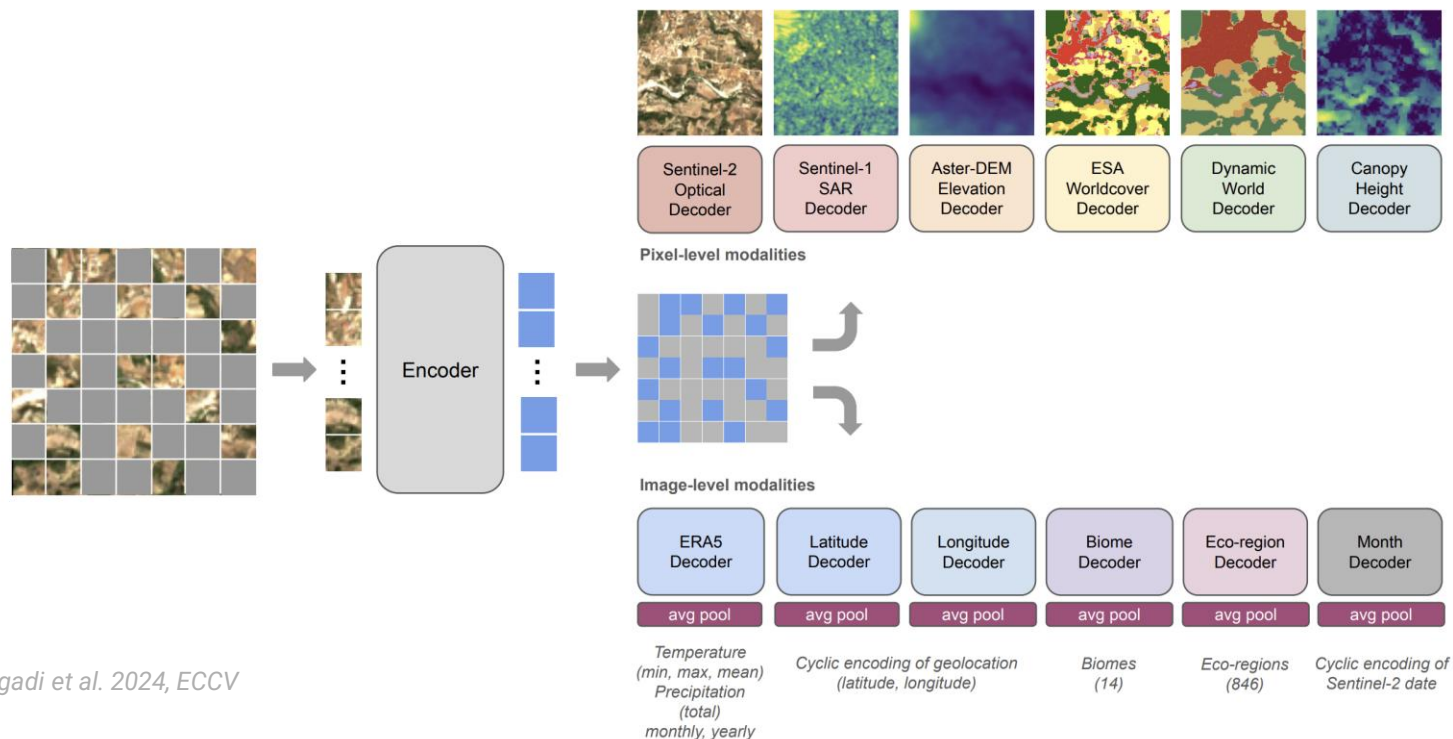


Self supervised deep learning has enabled the large-scale learning of representations optimized for general purpose use

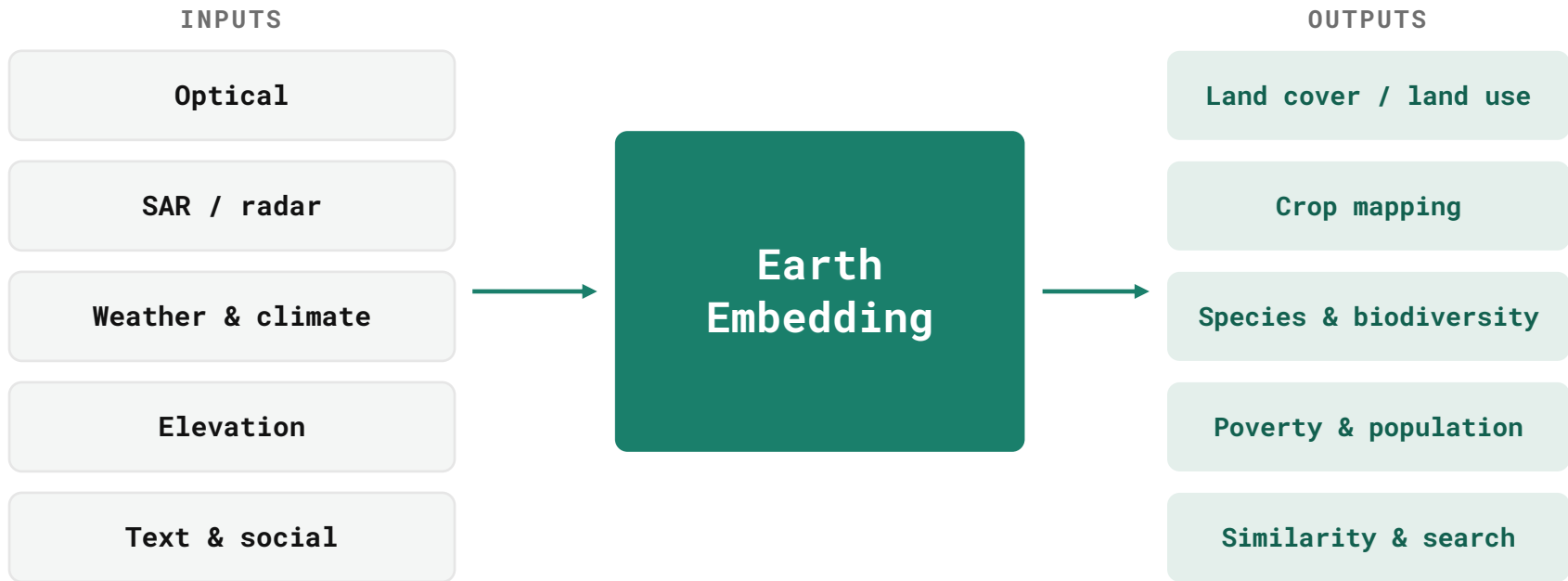
Masked Auto-Encoder (MAE)



Cross-modal Auto-Encoding for E0 data



Embeddings are a new remote sensing layer



Earth embeddings sit in an intermediate data layer between inputs and outputs

GeoFMs vs Earth embeddings

Geospatial foundation models

- Large multi-modal models (Clay, Prithvi, Olmo Earth, Alpha Earth)
- Can be fine-tuned for a specific task
- Architecture and learning objective decisions are made here

Earth embeddings

- The location-indexed vector outputs
- Can be stored, shared & queried on their own
- Often a by-product of GeoFMs – but often the part users actually touch!

A GeoFM is the factory; Earth embeddings are the product you take home.

PART 2

Earth Embedding Applications

What Earth embeddings unlock for practitioners

Presented by **Konstantin Klemmer**

$$E : (\text{lon}, \text{lat}, \text{time}) \rightarrow \text{emb} \in \mathbb{R}^d$$

Four ways people use embeddings

01

Direct prediction

Feed embeddings into a small model to map forest cover, population or species.

02

Geographic conditioning

Use embeddings as geo-priors that give your models location-specific context.

03

Synthesis & simulation

Condition generative models — a step toward simulators and digital twins.

04

Geo-semantic search

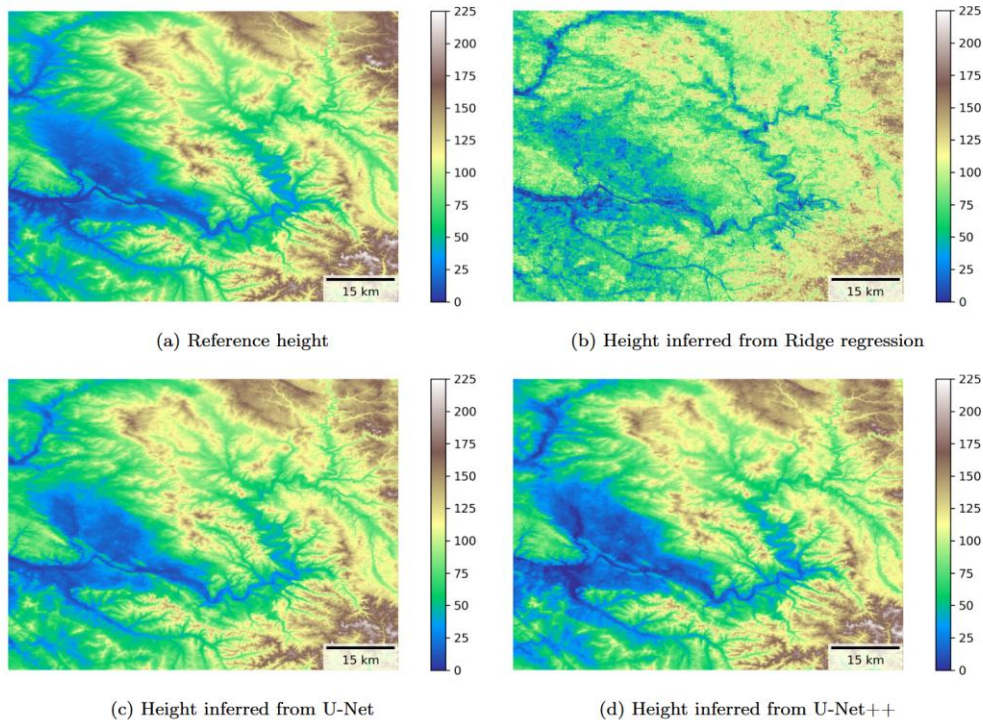
Find similar locations anywhere on Earth, fast — query by example or concept.

Direct prediction



- **Plug embeddings in where you'd use raw data.** Forest cover, population density, temperature, species presence.
- **Competitive with hand-crafted features.** Matches or beats specialist features with a random forest on tree-species ID.
- **Smaller models, less compute.** The heavy lifting is already baked into the embedding.
- **No raw-data pipeline.** Skip querying, downloading and pre-processing terabytes of imagery.

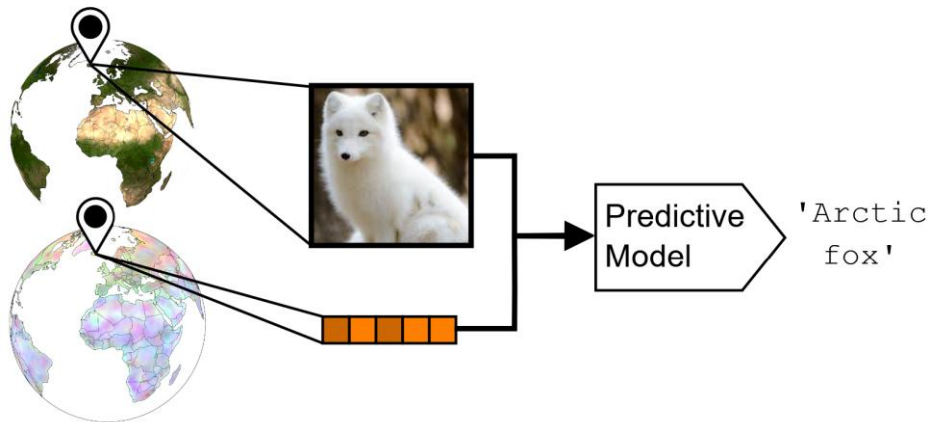
Predicting elevation from AEF embeddings



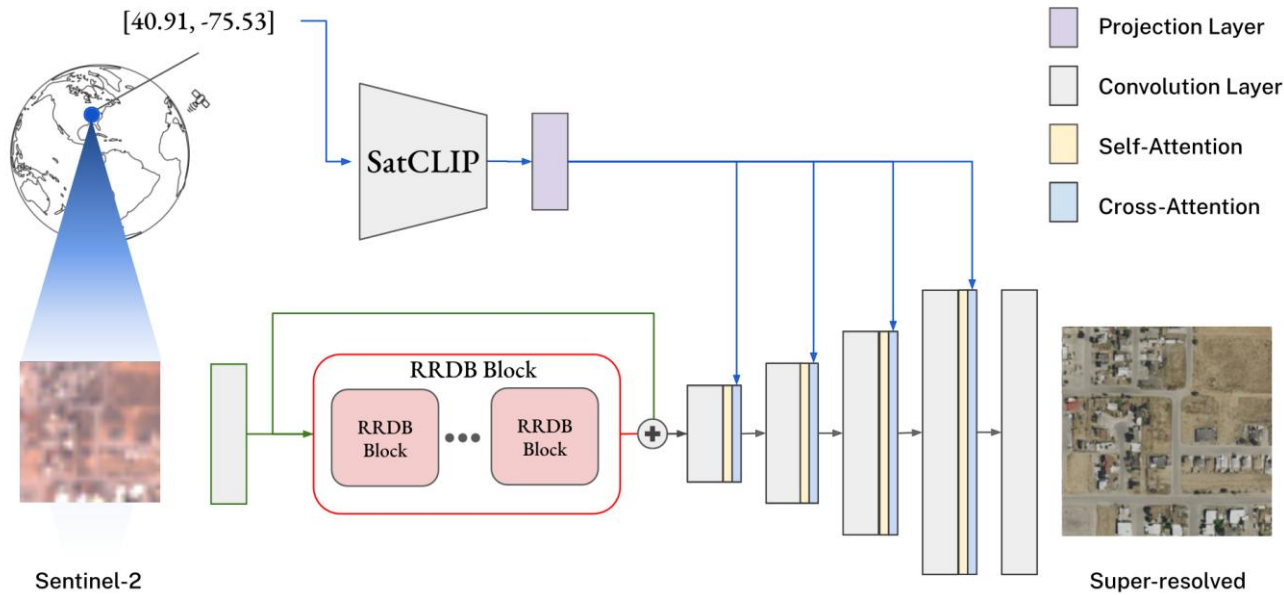
Hamoudzadeh et al. 2026, arXiv

Geographic conditioning (geo-priors)

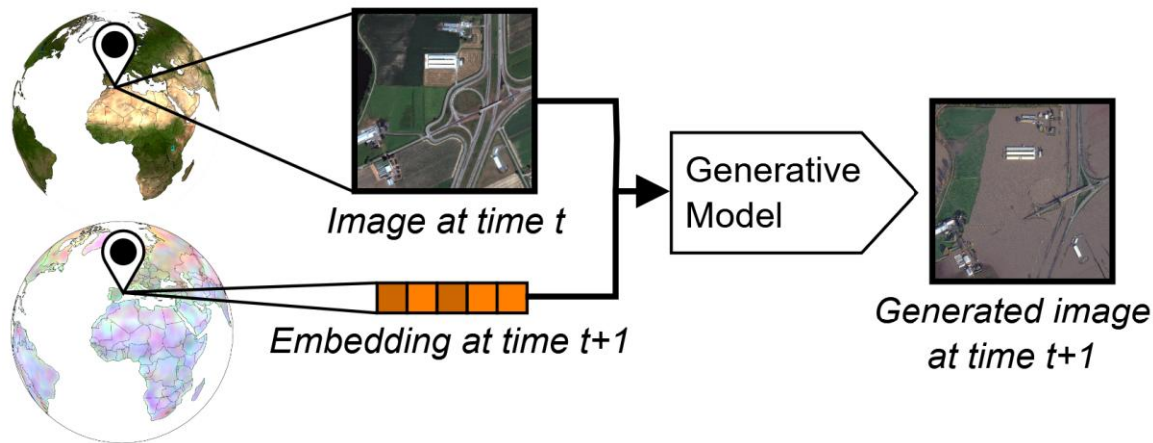
- **Add location context to any model.** Embeddings act as an auxiliary input — a geographic prior.
- **Better fine-grained classification.** Knowing “where” helps tell an Arctic fox from a similar-looking species.
- **Domain transfer across regions.** Helps a segmentation model trained in the USA work in Somalia.
- **Robust under limited labels & shift.** Extra context boosts performance when data is scarce.



Earth Embeddings improve image super-resolution

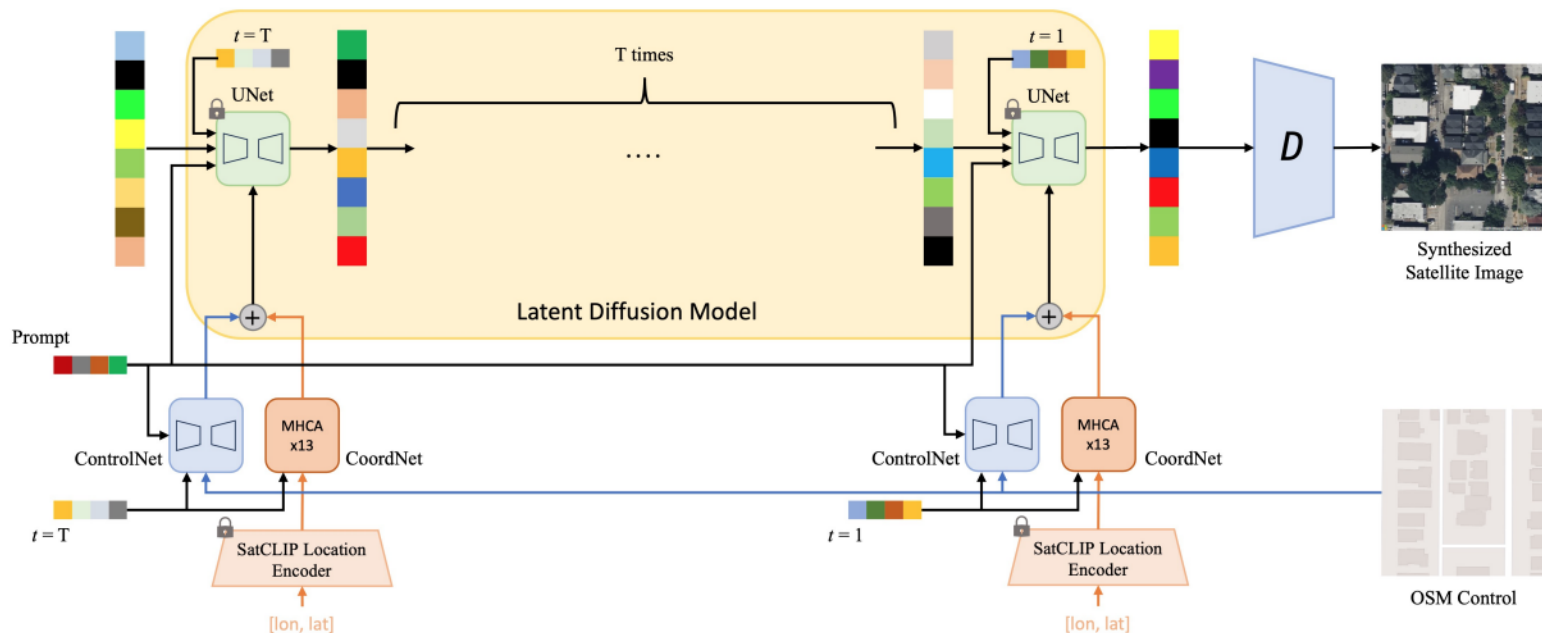


Data synthesis & simulation



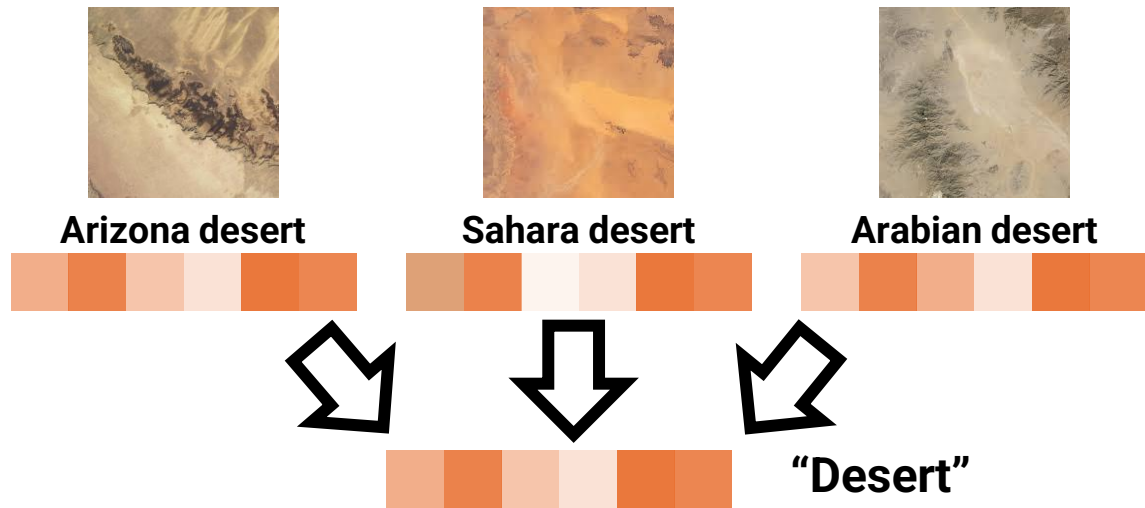
- **Condition image generators on location & time.** GeoSynth produces contextually-aware satellite imagery.
- **Toward Earth simulators.** Blend observations with physical knowledge — a path to digital twins.

Data synthesis & simulation



Fast, global geo-semantic search

Index the planet once, then ask — by example or by concept — “where else looks like this?”



- **Matching is just vector similarity — global results in seconds, no re-processing of imagery.** Build wall-to-wall similarity maps from a single reference point.



Feedback

lgnd.ai

✦ Find center pivot agriculture fields in California

✦ Find oil refineries in New Jersey

✦ Find new solar arrays in Texas

✦ Find new logging sites in Oregon

What can I help you discover?

Find... I



The payoff

What embeddings unlock

Fast global search

Query similar places worldwide in seconds — impossible with raw-imagery pipelines.

Universal geo-priors

Drop location context into any model, anywhere on Earth.

Accessible analytics

No GPU, no terabytes — a laptop and a linear model can go far.

From fragmented, task-specific GeoAI pipelines → to one coherent, reusable representation.

Your turn

You are part of this community

- **You don't need to be an ML expert.** If you can run a regression, you can use Earth embeddings.
- **Start from your own questions.** Bring the maps and variables your field already cares about.
- **Pre-computed embeddings are downloadable today.** AlphaEarth on Earth Engine, SatCLIP & others on GitHub / Hugging Face.
- **Demo 1 is next.** We'll go from an embedding to a real prediction together.

Up next → Interactive Demo 1: using Earth embeddings

INTERACTIVE

Demo 1 • Break

Hands-on: using Earth embeddings → 15-minute break

github.com/konstantinklemmer/isprs26-embeddings-tutorial

$$E : (\text{lon}, \text{lat}, \text{time}) \rightarrow \text{emb} \in \mathbb{R}^d$$

PART 3

Types of Earth embeddings

How embeddings are made, served, and chosen

Presented by **Anthony Fuller** · **Konstantin Klemmer**

$$E : (\text{lon}, \text{lat}, \text{time}) \rightarrow \text{emb} \in \mathbb{R}^d$$

Recap: What is an Earth embedding?

$$E : (\text{lon}, \text{lat}, \text{time}) \rightarrow \text{emb} \in \mathbb{R}^d$$

A family of embedding functions E maps a location in space and time to a d -dimensional vector. E comes in two forms:

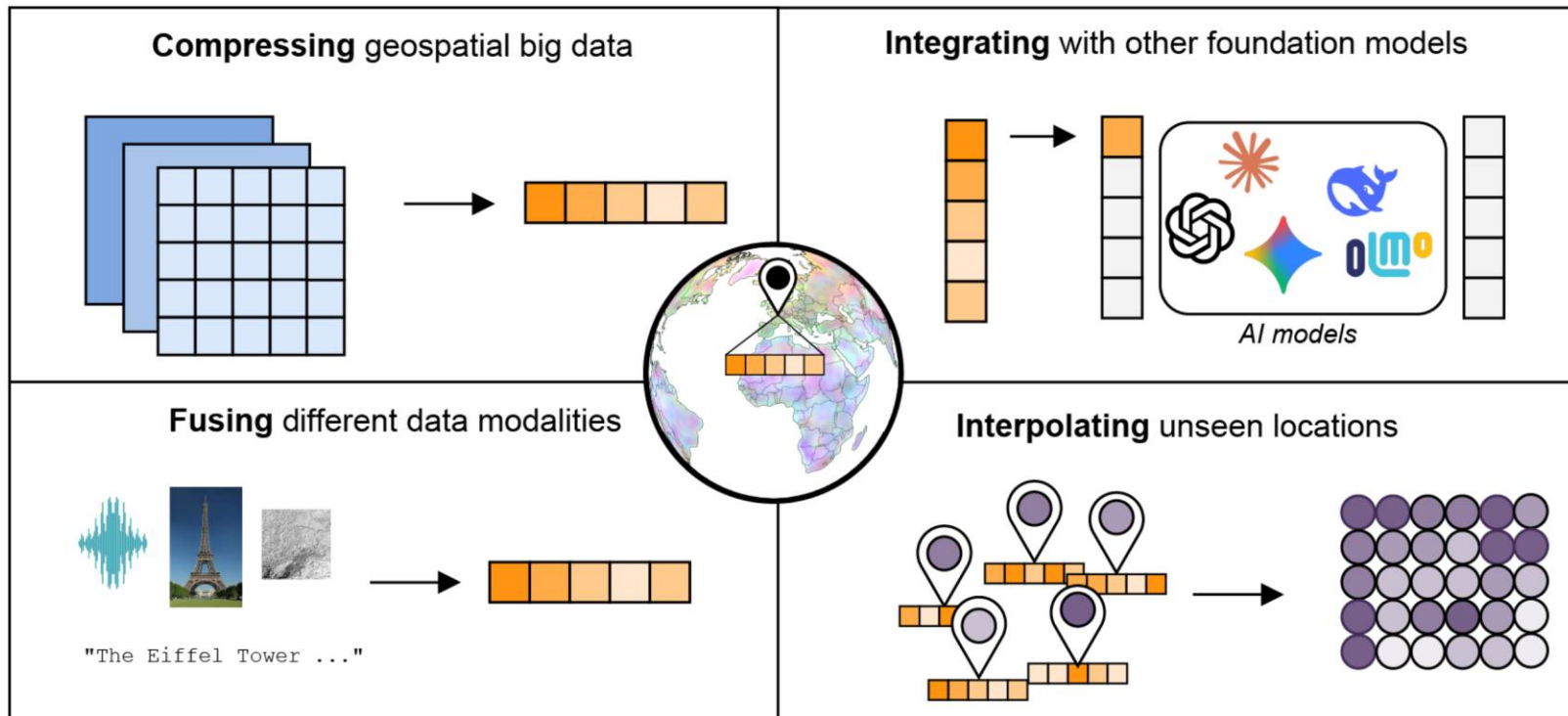
Explicit

emb from raw data (e.g. imagery)

Implicit

emb from location alone (INR)

Four functions of Earth embeddings



Two interfaces: model vs database

Embeddings as a MODEL

Ship code + weights; the user runs inference.

- Flexible — query anything, fine-tune
- Needs ML know-how, GPU, data

Embeddings as DATA

Ship a database of pre-computed vectors; the user looks them up.

- No ML or compute needed — just query
- Static — can't alter or inspect the model

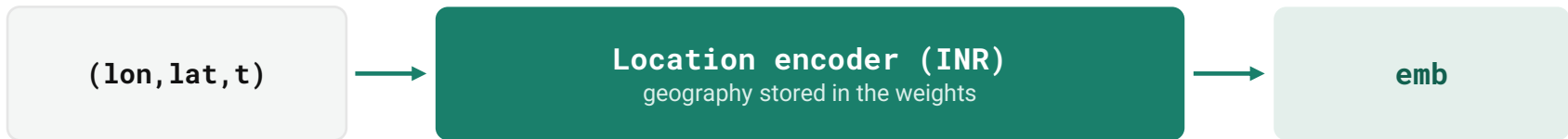
AlphaEarth, Tessera, Major TOM ship as data; SatCLIP, GeoCLIP, Climplicit ship as models.

Two ways to generate embeddings

EXPLICIT - feature extraction



IMPLICIT - neural representation (INR)



Explicit reads data at the location; implicit needs only the coordinates as inputs

PART 3.1

Direct (explicit) embeddings

Embed the raw data with a vision model

$$E : (\text{lon}, \text{lat}, \text{time}) \rightarrow \text{emb} \in \mathbb{R}^d$$

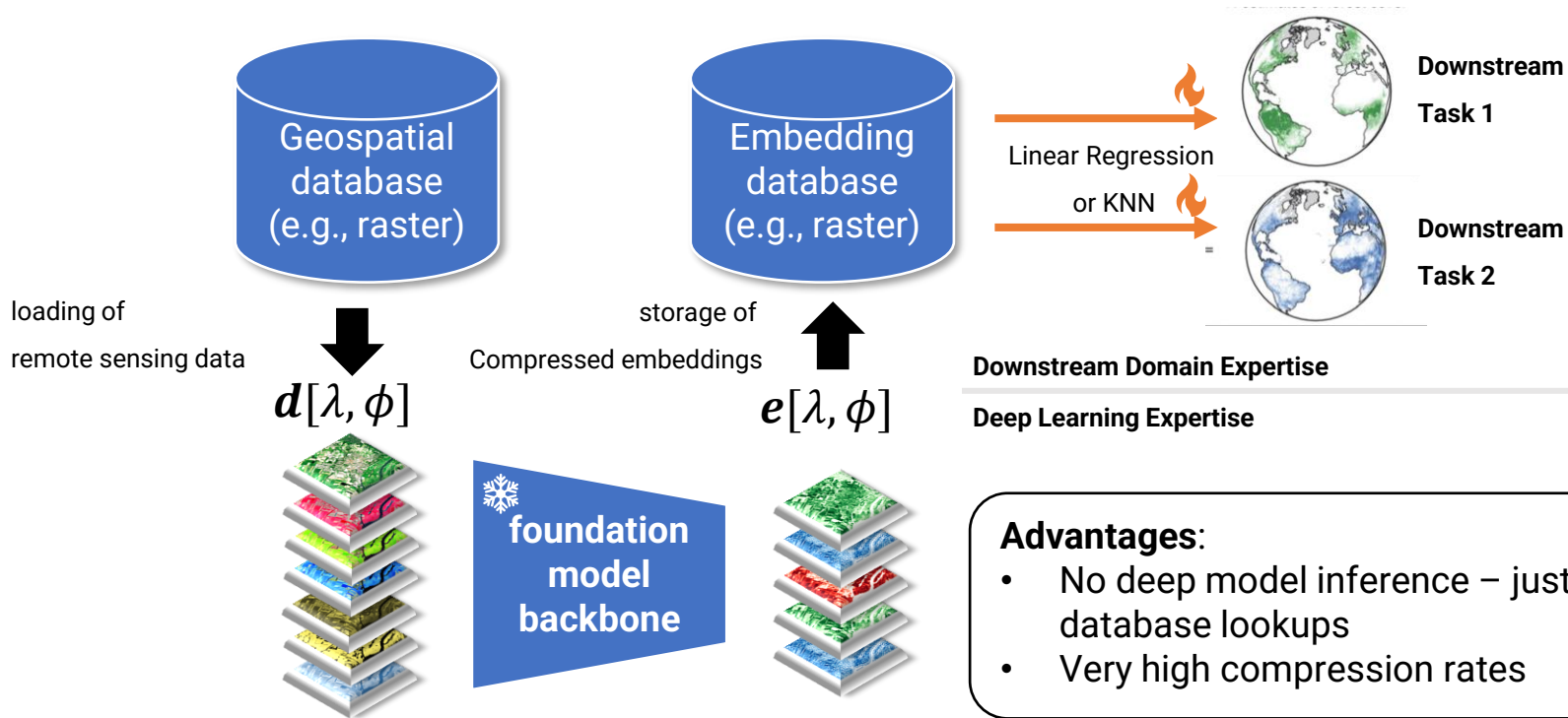
Embed the imagery directly

- **Run a vision model over geolocated data.** The embedding vector reflects the actual image content at that place & time.
- **High native resolution, fine-grained.** Great for search/retrieval and detailed mapping (land cover, crops).
- **Patch- or pixel-level outputs.** Whole-patch (Clay) or per-pixel (AlphaEarth, TESSERA) embeddings.
- **Powerful but heavier to run.** Often pre-computed and shipped as a database.

Examples

- MOSAIKS
- Clay
- MMEarth
- Galileo
- AlphaEarth

Explicit Embeddings



Inspired by Konrad Schindler's talk on Foundation Models @LPS 2025

Resolution and modality

Resolution

Patch-level (one vector per tile, e.g. Clay) vs pixel-level (one vector per pixel, e.g. AlphaEarth).
Coarse is cheap; fine localises detail.

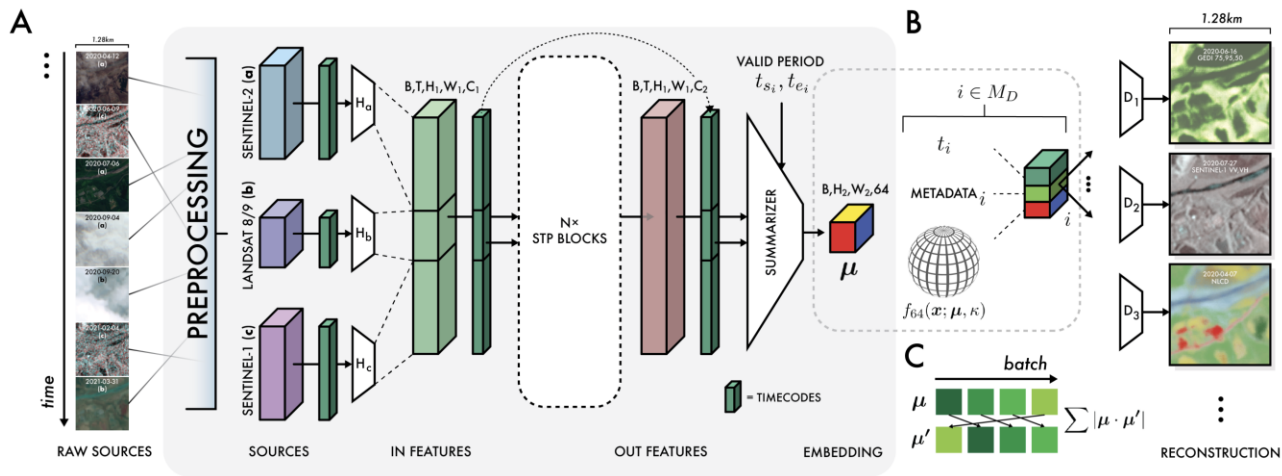
Modality

What sensors went in. SatCLIP is optical-only; AlphaEarth fuses more inputs — which can matter a lot for, say, crop mapping.

Match the embedding to the task

Mapping field boundaries? You want high resolution and the right input modalities. Modelling a broad climate gradient? Coarse, optical embeddings may be plenty — and far cheaper.

AlphaEarth Foundations

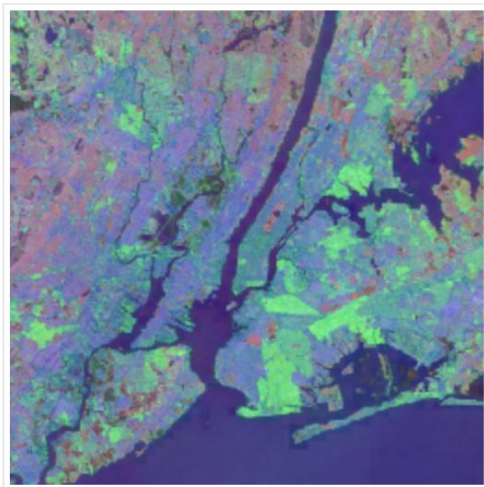


A pre-computed global embedding field on Google Earth Engine. Download the vectors – no model to run yourself.

- **Built for sparse-label mapping.** Add a light classifier on top and map at scale.

AEF is released as precomputed dataset

Satellite Embedding V1



Dataset Availability

2017-01-01T00:00:00Z–2024-01-01T00:00:00Z

Dataset Provider

[Google Earth Engine](#) [Google DeepMind](#)

Earth Engine Snippet

```
ee.ImageCollection("GOOGLE/SATELLITE_EMBEDDING/V  
1/ANNUAL") 
```

PART 3.2

Implicit embeddings (INRs)

Location information stored in the neural network's weights

$$E : (\text{lon}, \text{lat}, \text{time}) \rightarrow \text{emb} \in \mathbb{R}^d$$

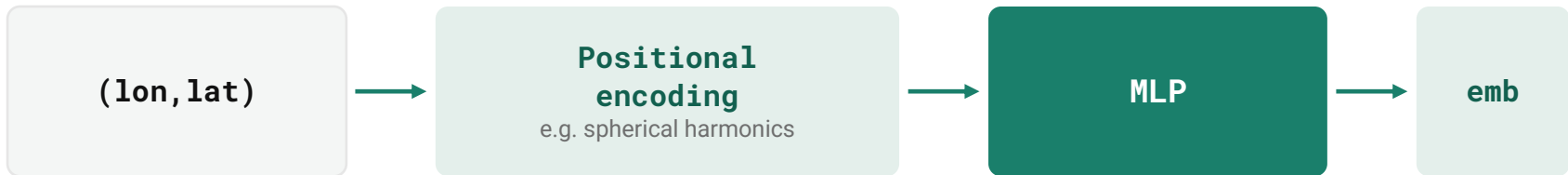
Encode location, not imagery

- **Input is just coordinates — no imagery at query time.** The model has already absorbed geographic info into its weights.
- **Location encoders.** A small network maps (lon, lat, time) straight to a vector.
- **Lightweight & queryable anywhere.** Smooth surface in space & time — it interpolates to unseen points.
- **Lower resolution, broad context.** Best for conditioning and large-scale prediction (for now!).

Examples

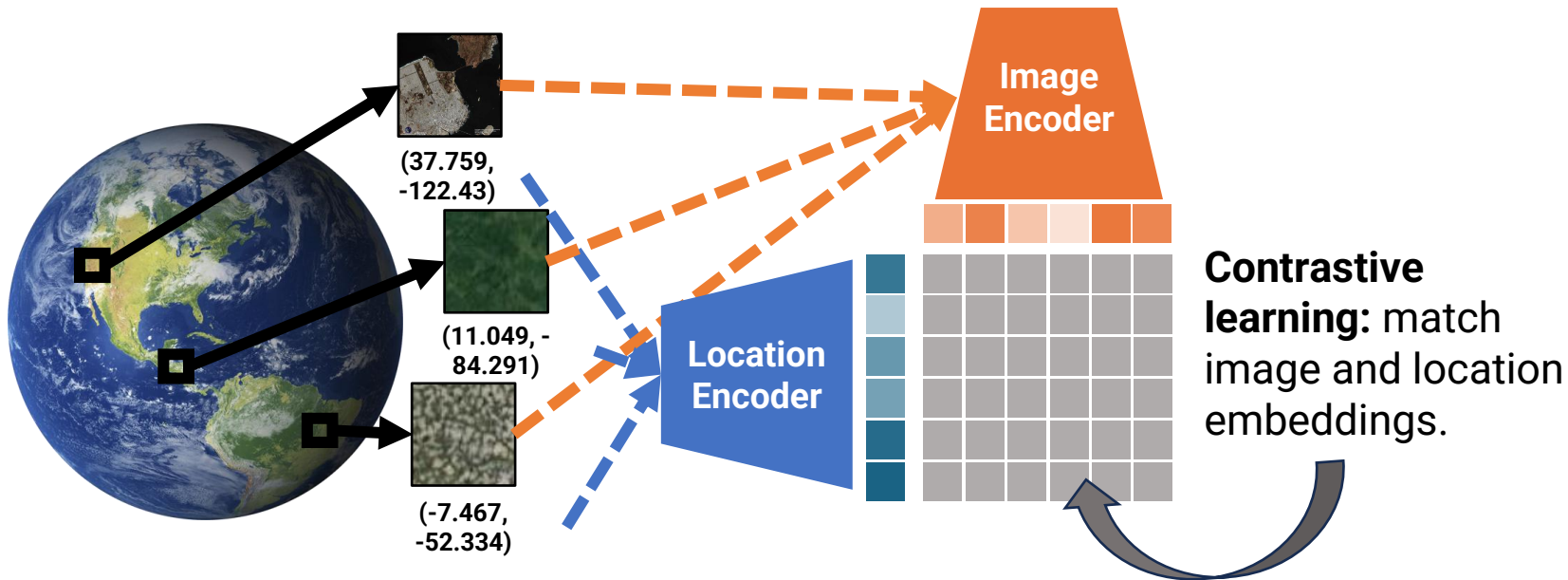
- SatCLIP
- GeoCLIP
- SINR
- Climplicit

Location encoders & spherical harmonics



- **Spherical harmonics are basis functions native to the sphere.** No distortion at the poles, unlike a flat lon/lat grid.
- **The network learns weights on these bases.** Any location on Earth becomes a smooth, learnable function — then interpolates between training points.

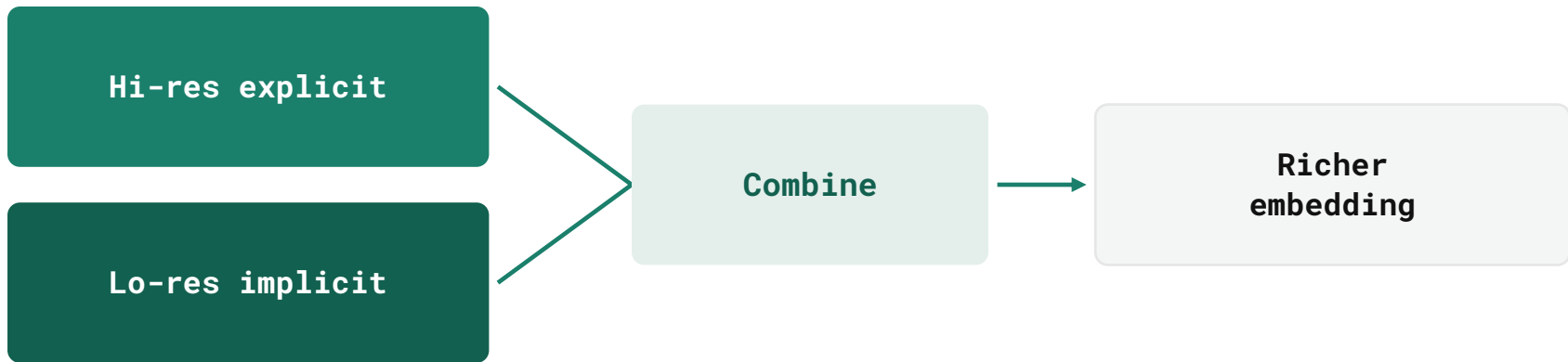
SatCLIP: learning places from images



Explicit vs implicit – the crux

	Explicit (direct)	Implicit (INR)
Input at query time	Imagery / raw data	Coordinates only
Resolution	High, fine-grained	Lower, broad
User compute	Heavier (or pre-computed)	Light
Interpolates?	No – needs data	Yes – smooth surface
Best for	Search, detailed mapping	Conditioning, large-scale
Usual interface	Database	Model
Examples	MOSAICS, Clay, AlphaEarth	SatCLIP, GeoCLIP, SINR

Hybrid approaches



- **Simplest: concatenate hi-res + lo-res vectors.** Won the NeucoBench challenge at CVPR EarthVision 2025.
- **Smarter: retrieval-augmented (RANGE).** Use the implicit embedding to fetch high-res features on demand.

Add better together

Which embedding should I use?

Need fine detail or search?

Go explicit (AlphaEarth, Clay). Higher resolution localises features.

Need any-point queries?

Go implicit / INR (SatCLIP, SINR). It interpolates to places with no data.

No GPU or ML pipeline?

Pick an embeddings-as-data product you can simply download and query.

Specific sensors matter?

Check the modalities — optical-only vs multi-modal can decide your task.

Rule of thumb: match resolution, modality and interface to your task — then start simple.

PART 3

Questions?

A few minutes for questions before the hands-on demo

$$E : (\text{lon}, \text{lat}, \text{time}) \rightarrow \text{emb} \in \mathbb{R}^d$$

INTERACTIVE

Demo 2

Make your own embeddings: MOSAIKS RCF · build a similarity map:

`github.com/konstantinklemmer/isprs26-embeddings-tutorial`

$$E : (\text{lon}, \text{lat}, \text{time}) \rightarrow \text{emb} \in \mathbb{R}^d$$

Challenges & opportunities ahead

Model capacity

GeoFMs are still small compared to state-of-the-art LLMs or VLMs

Data curation & scaling

What does one more place, time or sensor actually buy you?

Interpretability

Which part of a vector means “forest”? Still largely a black box.

Spatio-temporal heterogeneity

Geospatial data is differently complex in different areas and temporal intervals

Better evaluation

Benchmarks for interpolation, transfer & retrieval, not just image tasks

Toward digital twins

Can GeoFMs learn the physics of our planet from data alone?

An open, shared ecosystem

Openness comes in degrees:

Open-weight
weights only

Open-source
weights + code + data

Open-process
process & participation

The hidden cost is storage, not compute

Embedding all land on Earth costs only ~\$500 of GPU time — but storing the result runs to ~\$100,000 / year. Sustaining open embeddings is a community effort.

Resources & how to get involved

Tutorial code & notebooks

github.com/konstantinklemmer/isprs26-embeddings-tutorial

Readings

- Klemmer, Konstantin, et al. "Earth Embeddings: Towards AI-centric Representations of our Planet." IEEE GRSM (2026).
- Fang, Heng, et al. "Earth Embeddings as Products: Taxonomy, Ecosystem, and Standardized Access." arXiv (2026).
- Rolf, Esther, et al. "Mission Critical–Satellite Data is a Distinct Modality in Machine Learning." ICML (2024).
- Corley, Isaac, et al. "No One Knows the State of the Art in Geospatial Foundation Models." arXiv (2026).
- Betti, Livia, et al. "What's in an Earth Embedding? An Explainability Analysis of Location Encoders." arXiv (2026).
- Kaur, Amandeep, et al. "Pretrain Where? Investigating How Pretraining Data Diversity Impacts Geospatial Foundation Model Performance." CVPR (2026).
- van der Plas, Thijs L., et al. "Better Together: Evaluating the Complementarity of Earth Embedding Models." arXiv (2026).

Stay in touch

Join the TorchGeo community!



github.com/torchgeo

Thanks for joining us on this first orbit.

github.com/konstantinklemmer/isprs26-embeddings-tutorial



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